

Fluid Blob Background Effect

Interactive cursor-tracking gradient orb for portfolio hero sections

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Reference	Dave Lee portfolio (Inelase.com) — Fluid blob hero effect
Platform	Web (Responsive) — Desktop & Mobile

1. Overview

This document specifies the requirements for a fluid, animated gradient blob effect to be implemented on the hero section of a personal portfolio website. The effect consists of a soft-edged, multi-color radial gradient orb that follows the user's cursor across the viewport, organically morphing its shape over time and emitting subtle ripple-like distortion waves. The visual result is an ambient, almost living background element that adds depth and personality without distracting from the primary content.

The reference implementation was observed on Dave Lee's portfolio and represents a state-of-the-art approach to interactive hero backgrounds currently popular among top-tier UX/product designers.

2. Goals & Non-Goals

2.1 Goals

- Deliver an ambient, cursor-reactive blob that enhances visual interest on the portfolio hero section.
- Create a smooth, 60fps animation that feels organic and premium, not mechanical.
- Ensure the effect is non-intrusive — all text content remains fully legible at all times.

- Support light mode with a soft white/light-grey background where the blob blends naturally.
- The blob must follow the mouse with a smooth lag/inertia, not snapping instantly to cursor position.
- Include idle animation so the blob remains alive even without user interaction.
- Support graceful degradation on low-end devices and mobile (reduced motion / static fallback).

2.2 Non-Goals

- Dark mode variant is out of scope for v1.
 - The blob should NOT render on any page other than the portfolio hero/landing section.
 - This is not a full WebGL particle system — no 3D geometry or depth effects.
 - No user controls or settings UI is required.
 - The blob should not be clickable or interactive beyond cursor tracking.
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3. Visual Design Specification

3.1 Blob Anatomy

The blob is composed of layered semi-transparent radial gradient ellipses rendered via canvas or CSS/SVG. From the reference video, the following visual layers were observed:

- **Outer glow:** Outer glow layer:
 - Very large, extremely low-opacity diffuse halo (opacity ~5–10%) in cool blue-grey.
 - Radius approximately 30–40% of the viewport short edge.
- **Mid body:** Mid body layer:
 - Medium-opacity (30–50%) radial gradient from steel blue / periwinkle (#6B8FD4, #8AA8E8) to transparent.
 - Radius approximately 18–25% of viewport.
- **Core highlight:** Core highlight:
 - Small, brighter, slightly higher-opacity center (opacity ~60%) in light blue (#A8C4F0).
 - Creates the illusion of a light source or depth within the blob.
- **Ripple rings:** Ripple / distortion rings:
 - Concentric low-opacity rings (opacity ~3–8%) radiating outward from the blob center.
 - Animate outward at ~0.5–1.5 px/frame, fading to zero before edge.
 - Spacing between rings ~60–100px. Emit new ring every 1.5–2 seconds.

3.2 Color Palette

Role	Hex Value	Usage
Blob Core	#A8C4F0	Bright highlight center
Blob Mid	#6B8FD4 → #8AA8E8	Steel blue to periwinkle body
Outer Halo	#B8CDE8 (5% opacity)	Cool blue-grey diffuse glow
Ripple Rings	#6B8FD4 (3–8% opacity)	Expanding concentric rings
Page Background	#F5F5F3 / #FFFFFF	Off-white / paper hero background

3.3 Size & Positioning

- Default spawn position: upper-right quadrant of the hero viewport (~65% from left, ~30% from top).
- Initial blob diameter: ~350–450px on a 1440px viewport (scales proportionally to viewport width).
- Blob positioning: absolute, z-index behind text content (z-index: 0), text at z-index: 10+.
- Pointer-events must be set to none on the blob canvas so it never interferes with text selection or links.

4. Animation Specification

4.1 Cursor Tracking — Inertia System

The blob does NOT snap to cursor position. It follows with smooth exponential lerp (linear interpolation):

```
blobX += (targetX - blobX) * LERP_FACTOR // LERP_FACTOR = 0.06
(recommended)
```

- LERP_FACTOR range: 0.04 (very sluggish, dreamy) to 0.10 (snappier, still smooth). Default: 0.06.
- Apply lerp every requestAnimationFrame tick (targeting 60fps).
- The blob center tracks the cursor. The outer halo position may have additional secondary lag (~0.03 factor) for extra depth.

4.2 Idle / Autonomous Animation

When the cursor is stationary or has left the viewport, the blob continues to breathe and drift:

- Drift: apply slow sinusoidal offset to X and Y using time-based sin/cos with different frequencies.
 - X drift: amplitude 30–60px, period ~8–12 seconds
 - Y drift: amplitude 20–40px, period ~10–15 seconds (different phase from X to avoid circular orbit)
- Scale pulse: blob radius oscillates $\pm 5\text{--}8\%$ of its base size using $\sin(\text{time} * 0.5)$. Period ~4–6 seconds.
- Color shift: very subtle hue rotation ($\pm 8^\circ$) over 20-second cycle for iridescent quality.

4.3 Morphing Shape

The blob is not a perfect circle. Its edges organically distort using a noise-based displacement:

- Use simplex noise (or equivalent) applied radially around the blob boundary to offset edge points by $\pm 5\text{--}15\%$ of the radius.
- Noise evolves slowly over time (time offset ± 0.003 per frame) creating organic movement.
- Shape update frequency: every frame (60fps) — must be smooth with no visible jumps.
- The blob should never collapse to a sharp or geometric shape.

4.4 Ripple / Distortion Rings

Concentric expanding rings emanate from the blob center at regular intervals:

- New ripple emitted every 1,500–2,000ms.
- Each ring starts at blob core radius and expands to $\sim 3\times$ the blob radius before fading out.
- Opacity animation: starts at 5%, peaks at 8% at mid-expansion, fades to 0% at full radius.
- Ring stroke width: 1–1.5px; color matches blob mid-color at very low opacity.
- Maximum concurrent rings: 4. Oldest ring is removed when new one is spawned at max count.
- Ring expansion speed: 0.8–1.2px per frame.

4.5 Page Entry Animation

- On page load, blob fades in over 1,200ms with ease-out.
 - Simultaneously scales from 60% to 100% of final size (origin: blob center).
 - Initial position: default upper-right spawn position (not cursor) until first mousemove event.
 - On first mousemove, blob begins tracking cursor position using lerp from its current location.
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5. Technical Requirements

5.1 Functional Requirements

ID	Requirement	Pri	Notes
FR-01	Blob canvas renders behind all page content without blocking pointer events.	P0	pointer-events: none; z-index layering required.
FR-02	Blob follows cursor with lerp inertia at 60fps via requestAnimationFrame.	P0	LERP_FACTOR = 0.06 default.
FR-03	Blob renders using HTML5 Canvas 2D or WebGL (Canvas preferred for compatibility).	P0	Canvas approach recommended.
FR-04	Idle drift animation active when cursor is not moving or is off-screen.	P1	Sin/cos time-based offsets.
FR-05	Shape morphing via simplex noise applied to blob boundary.	P1	3rd party noise library permitted.
FR-06	Ripple rings emit from blob center at 1.5–2s intervals.	P1	Max 4 concurrent rings.
FR-07	Blob fades and scales in on page load over 1,200ms.	P1	CSS opacity + JS scale.
FR-08	Canvas resizes correctly on viewport resize (debounced, 200ms).	P0	Re-initialize canvas dimensions.
FR-09	prefers-reduced-motion media query disables all animation, shows static blob.	P0	Accessibility requirement.
FR-10	Mobile: blob tracks touch position or displays static idle animation.	P2	Touch events or static fallback.
FR-11	No external image assets required — all rendering is programmatic.	P1	Pure canvas / CSS only.
FR-12	Effect is scoped strictly to the hero section container.	P0	Canvas clipped to hero bounds.

5.2 Performance Requirements

Metric	Target	Constraint
Animation frame rate	60fps sustained	No dropped frames on modern hardware
JS bundle overhead	< 8KB gzipped (excl. noise lib)	Noise lib ~3KB, total < 12KB acceptable
CPU usage (idle tab)	< 2% CPU when tab unfocused	Use document.visibilityState to pause rAF

Metric	Target	Constraint
First Contentful Paint impact	Zero impact — load deferred	Init after DOMContentLoaded or requestIdleCallback

6. Implementation Guide

6.1 Recommended Architecture

The blob effect should be implemented as a self-contained ES module (FluidBlob.js) with zero runtime dependencies except for an optional noise library. The canvas element is injected into the hero container and all cleanup is handled via a destroy() method for SPA route changes.

6.1.1 Core Module Structure

```
FluidBlob.init(containerEl, options) // Mount and start
FluidBlob.destroy()                 // Cleanup for SPA
FluidBlob.setTheme(palette)         // Future: theme switching
```

6.2 Recommended Tech Stack

- Vanilla JS (ES2020+) — no framework dependency. Works with React, Next.js, Astro, Svelte, or plain HTML.
- simplex-noise (npm: ~3KB gzipped) for organic shape morphing. Alternatively: hand-rolled Perlin noise.
- HTML5 Canvas 2D API for rendering. WebGL is not required for this effect.
- requestAnimationFrame loop with time delta for frame-rate-independent animation.
- ResizeObserver (not window resize) to respond to container size changes.

6.3 Canvas Rendering Order (per frame)

Execute the following draw operations in order each frame to achieve correct layering:

1. clearRect — clear entire canvas
2. Draw ripple rings (oldest first, outer first so new ones appear on top)
3. Draw outer halo gradient (largest, most transparent)
4. Draw mid body gradient
5. Draw core highlight (smallest, highest opacity)
6. Apply canvas globalCompositeOperation: 'screen' or 'normal' depending on background

6.4 Configurable Parameters (options object)

Parameter	Default	Range	Description
<code>lerpFactor</code>	0.06	0.02–0.15	Cursor follow speed. Lower = dreamier.
<code>blobRadius</code>	200	100–400px	Base blob radius at 1440px viewport.
<code>haloScale</code>	2.2	1.5–3.0×	Outer halo size relative to blobRadius.
<code>noiseSpeed</code>	0.003	0.001–0.01	How fast the shape morphs. Lower = slower.
<code>noiseStrength</code>	0.12	0.05–0.25	Edge morphing intensity (% of radius).
<code>rippleInterval</code>	1800	1000–3000ms	Milliseconds between new ripple rings.
<code>rippleMaxCount</code>	4	2–6	Max concurrent ripple rings.
<code>driftAmplitudeX</code>	40	0–100px	Idle horizontal drift amplitude.
<code>driftAmplitudeY</code>	30	0–80px	Idle vertical drift amplitude.
<code>colorCore</code>	#A8C4F0	any hex	Blob core highlight color.
<code>colorMid</code>	#6B8FD4	any hex	Blob body color.

7. Accessibility

- **Reduced motion:** prefers-reduced-motion: When this media query is active, all blob animation must cease. The blob may be rendered as a static gradient at its default position, or omitted entirely.
- **ARIA:** The canvas element must have `aria-hidden="true"` and `role="presentation"` — it carries no semantic information.
- **Contrast:** Text contrast: The blob must never reduce text contrast below WCAG AA (4.5:1 ratio). Since the blob is always on a light background and text is dark, this should be inherently satisfied, but must be verified.
- **Battery:** Battery-conscious: Use `document.visibilityState` and the Page Visibility API to pause the animation loop when the tab is not visible.

8. Testing & Acceptance Criteria

8.1 Visual QA Checklist

- Blob smoothly follows cursor with no jittering or snapping.
- Blob continues idle drift and breathing when cursor is still for >3 seconds.
- Ripple rings expand from blob center, fade out before edge of blob halo.
- Shape appears organic and non-circular at all times.

- All page text remains fully readable above the blob.
- On viewport resize, blob canvas resizes correctly within 250ms.
- Entry animation plays correctly on cold page load.

8.2 Performance Tests

- Chrome DevTools Performance tab: no red frames in the rAF loop during 10s cursor-tracking test.
- FPS counter remains at 58–60fps on MacBook Air M-series, Intel i5 (2020+), and equivalent.
- Tab unfocused: CPU drops to near-zero (animation loop paused).
- Lighthouse performance score not impacted by more than 2 points vs baseline (no blob).

8.3 Accessibility Tests

- Enable prefers-reduced-motion in OS settings: blob becomes static, no animation plays.
- Screen reader announces no blob-related content (aria-hidden verified in DOM inspector).

8.4 Browser & Device Support

Environment	Support Level	Notes
Chrome 100+, Edge 100+	Full	Primary target
Firefox 100+	Full	Test ripple rendering
Safari 15+	Full	Test canvas perf on M-chip
iOS Safari 15+	Touch tracking or static	Touch events required
Android Chrome	Touch tracking or static	Performance varies by device

9. Open Questions

#	Question	Owner / Target
1	Should the blob color palette adapt to match the section content or portfolio brand color?	Designer — before dev starts
2	Is a dark mode variant required for v1, or strictly out of scope?	Owner — v1 planning

#	Question	Owner / Target
3	What happens on touch/mobile: touch tracking, gyroscope idle, or fully static?	Engineer — mobile pass
4	Should the blob appear on project detail pages or strictly the hero/landing section?	Owner — scope decision
5	Is simplex-noise an acceptable dependency, or must everything be zero-dep?	Engineer — architecture review

10. Milestones & Delivery

Milestone	Deliverable	Estimated Duration
M1 — Proof of Concept	Static blob renders correctly on canvas, correct colors, sizes, no animation.	0.5 day
M2 — Cursor Tracking	Lerp-based cursor follow working at 60fps. Canvas z-layering correct.	0.5 day
M3 — Idle Animation	Drift, breathing, and color shift active during idle state.	0.5 day
M4 — Shape Morphing	Noise-based edge distortion rendering naturally.	1 day
M5 — Ripple Rings	Ripple ring system working with correct opacity fade and timing.	0.5 day
M6 — Polish & Entry	Page entry animation, performance optimization, reduced-motion support.	0.5 day
M7 — QA & Ship	Cross-browser testing, Lighthouse audit, accessibility check, deploy.	1 day

Total Estimated Engineering Time: 4.5 days | Document Version 1.0.0 | February 17, 2026